Use the shared card-voting rules: each voter chooses the higher of the two available letters. The first winner faces the waiting letter. Cards stay unchanged.

Problem 1: Can changing the waiting letter change the knockout winner for each set of cards? Explain how you can tell without playing every knockout.

V1
1st A
2nd B
3rd C

V2
1st A
2nd C
3rd B

V3
1st B
2nd A
3rd C

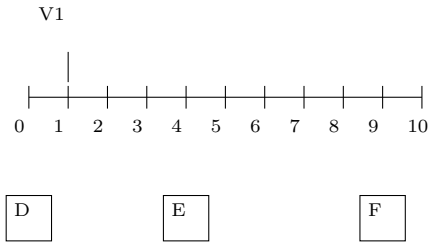
V1
1st A
2nd C
3rd B

V2
1st C
2nd B
3rd A

V3
1st B
2nd A
3rd C

A nearer letter ranks higher; equal distances make a tie, not a strict ranking. In this problem letters and favorites may be anywhere on a straight line.

Fixed locations



From the same favorite

D: 1 step
E: 3 steps
F: 8 steps

V1
1st D
2nd E
3rd F

Problem 2: Could these three cards come from distance choices on one line? Put A, B and C at different places and choose a favorite for each voter, or explain why it cannot be done.

V1
1st A
2nd B
3rd C

V2
1st B
2nd C
3rd A

V3
1st C
2nd A
3rd B

